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Music Artists & Albums Public Perception

Systems Engineering Curriculum
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Course: Programming for Data Analysis

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1 Project Overview

1.1 Title and Topic Angle

"Music Artists & Albums Public Perception", this project focuses on analyzing data related to how people perceive music, artists, albums, and musicals trends. To understand public perception of aspects such as releases or artists, enabling entities interested in the music market to make informed decisions based on real data from active communities.

1.2 Why is relevant?

It is relevant because singers, producers, and music companies can analyze public opinions about albums, songs, and artist interactions. With this information, they can better understand the interests and preferences of the audience. Also, they can identify what type of music people like the most and how trends change over time. This helps them make better decisions when creating new music or promoting their work.

1.3 Goal

The goal of this project is to understand music trends based on user opinions. We want to analyze how people talk about songs, artists, and albums. Another goal is to find patterns in the data that can help us predict future trends. This means we try to identify what could be popular in the next important moment in the music industry. To this end, we propose building an end-to-end data pipeline that collects, processes, and analyzes music review texts, artist metadata, and listener data to generate an interactive opinion analysis dashboard.

1.4 Scope and Limitations

We will only analyze data in English. This helps us keep the data consistent and easier to compare. If we include many languages, the analysis could become more complex and less accurate.

We will use data from two main sources: Reddit and Last.fm. Reddit provides opinions and discussions, while Last.fm provides listening behavior, such as most played artists, albums, and songs. This combination helps us understand both what people say and what they actually listen to.

However, this also has limitations. Not all users use Last.fm, so the listening data may not represent all audiences. Also, Reddit users may have different tastes compared to users on other platforms.

Another limitation is that we only collect a limited amount of data, so the dataset is relatively small. Finally, some comments or data may be unclear or not useful, which can affect the results of the analysis.

2 Data Sources

2.1 API Source: Last.fm

The API used to obtain the first data sample comes from the Last.fm platform and shares the same name. Its services consist of music tracking and a social network that creates detailed profiles of users listening habits by connecting to platforms such as Spotify or YouTube Music. Specifically, its API

is used to access this data, enabling the creation of applications, the generation of recommendations, personalized statistics, and the retrieval of artist information. To access the API service, we must register and then fill out a form specifying how we intend to use the service. This generates a key that allows us to retrieve data. Similarly, the following technical specifications were used to retrieve the data.

- **Base URL:** `https://ws.audioscrobbler.com/2.0/`.
- **Authentication Method:** API Key (passed as a parameter in the GET request).
- **Available Endpoints:** `chart.gettopartists`, `artist.getinfo`, `album.gettoptags`, and `track.search`.
- **Data Format:** JSON (standardized output for the Bronze layer).
- **Rate Limits:** 5 requests per second per IP address; non-commercial use only.
- **Key Fields for Analysis:** `artist.name`, `tags`, `playcount`, `listeners`, and `bio.content` (for initial NLP testing).

2.2 Web Scraping Source: Reddit

This Python script extracts posts and comments from Reddit, specifically from a subreddit named r/musicsuggestions.

The main objective is to collect music-related opinions from post titles and comments, which can later be used for analysis with sentiment analysis

2.3 Justification Selection

Reddit was selected as the primary data source due to its large volume of user-generated content and active discussion communities. Subreddits like r/musicsuggestions provide organic opinions and recommendations about music. The platform offers a public API, enabling easy and efficient data extraction without complex authentication. Data is structured (posts, scores, comments), which simplifies processing and analysis. It reflects real user preferences, making it valuable for sentiment analysis and trend detection. Additionally, content is continuously updated, ensuring relevance. The diversity of users increases variability and richness of the dataset. Overall, it is a practical and accessible source for text mining and opinion analysis tasks.

3 Data Relevance and Justification

Reddit provides highly relevant data for this project because it contains real, user-generated opinions about music in active communities like r/musicsuggestions. The extracted titles and comments directly reflect user preferences, recommendations, and discussions, aligning with the objective of analyzing music-related opinions. The inclusion of post scores adds a relevance metric, helping identify more influential or popular content. Comments provide richer context and qualitative insights beyond simple metadata. The data is timely and continuously updated, ensuring current trends are captured. Its semi-structured format (JSON) makes it suitable for preprocessing and analysis. Additionally, filtering (e.g., keywords) allows focusing on specific topics of interest. Therefore, Reddit is both a relevant and reliable source for text mining and sentiment analysis in this domain.

3.1 Complement API and Web Scraping

- This project combines two data collection methods: API and web scraping.
- The API *Last.fm* provides structured and quantitative data, such as artists, albums, and number of plays.
- This type of data is useful to understand listening behavior and popularity.
- On the other hand, web scraping from Reddit provides qualitative data, such as opinions, comments, and recommendations.
- This helps to understand how users feel and what they think about music.
- By combining both sources, we can analyze both behavior (what users listen to) and perception (what users say).
- This complement improves the quality of the analysis and provides a more complete view of music trends.
- It also helps to validate results by comparing data from different sources.

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4 Data Sample

4.1 API Sample

The following JSON code snippet represents a partial response from the `chart.gettopartists` endpoint, which is a method that retrieves artist data. In this example, it confirms the availability of text fields, such as artist names, and quantitative metadata, such as the number of listeners:

```
\end\begin{lstlisting}[language=json, caption={Last.fm API Response Sample}]
{
  "artists": {
    "artist": [
      {
        "name": "The Weeknd",
        "playcount": "385642100",
        "listeners": "3541200",
        "url": "https://www.last.fm/music/The+Weeknd"
      },
      {
        "name": "Taylor Swift",
        "playcount": "942156000",
        "listeners": "4215800"
      }
    ]
  }
```

```
}
```


```
"artists": {
    "artist": [
        {
            "name": "PinkPantheress",
            "playcount": "312041059",
            "listeners": "2983144",
            "mbid": "7441014f-f8f5-494f-81db-ff166fbc078d",
            "url": "https://www.last.fm/music/PinkPantheress",
            "streamable": "0",
            "image": [
                {
                    "#text": "https://lastfm.freetls.fastly.net/i/u/34s/2a96cbd8b46e442fc41c2b86b821562f.png",
                    "size": "small"
                },
                {
                    "#text": "https://lastfm.freetls.fastly.net/i/u/64s/2a96cbd8b46e442fc41c2b86b821562f.png",
                    "size": "medium"
                },
                {
                    "#text": "https://lastfm.freetls.fastly.net/i/u/174s/2a96cbd8b46e442fc41c2b86b821562f.png",
                    "size": "large"
                },
                {
                    "#text": "https://lastfm.freetls.fastly.net/i/u/300x300/2a96cbd8b46e442fc41c2b86b821562f.png",
                    "size": "extralarge"
                },
                {
                    "#text": "https://lastfm.freetls.fastly.net/i/u/300x300/2a96cbd8b46e442fc41c2b86b821562f.png",
                    "size": "mega"
                }
            ]
        },
        {
            "name": "The Weeknd",
            "playcount": "1103626666",
            "listeners": "5257261",
            "mbid": "c8b03190-306c-4120-bb0b-6f2ebfc06ea9",
            "url": "https://www.last.fm/music/The+Weeknd",
            "streamable": "0",
            "image": [
                {
```


Figure 1: Example of data obtained from API Last.fm

4.2 Scraping sample

We implement the web scraping with reddit because is a good tool to extract sentimental data, qualitative data, data that is more complex to analyze. Ver Figura 2


```
[
    {
        "title": "Recommendations",
        "score": 2,
        "comments": [
            "Judas Priest - \"Electric Eye\"",
            "Electric Eyes - Client Liason\n\nIt Took Me So Long To Get Here But Here I Am - KT Tunstall\n\nInto Th
            "[OpasK - CrusH](https://open.spotify.com/album/1uNXyoWMDTCejYixzNhzyj?si=qeNESmgVSqCLDC2GD940iQ)",
            "Electrify (band)",
            "[Digital Security Spotify ](https://open.spotify.com/track/3yrHfH4Ag417ZXqmomeObf?si=sF9sC8KbRMCiZHgsX
        ]
    },
```


Figure 2: Example of data obtained from Reddit

4.3 Data Quality and Cleaning

Some important point about web scraping is that we need to select the correct subreddit. If we only use a general subreddit like r/music, not all users talk about music reviews. Many posts are about news, events, or general discussions about artists.

Because of this, the data may not be useful for our objective. For example, we want opinions, recommendations, and user experiences about music. That is why a subreddit like r/musicsuggestions is more appropriate.

5 Data Architecture Overview

The project architecture follows the "Medallion Architecture" design pattern, ensuring data integrity, cleaning, and progressive aggregation of musical data from initial capture to final visualization. The flow is divided into three logical layers:

5.1 Bronze Layer

This layer acts as the landing zone for raw data originating from the Last.fm API and musical community scraping of Reddit.

- **Format:** Data is persisted in its original state within JSON files.
- **Nomenclature:** Files follow the standard `source_topic_YYYYMMDD_HHMMSS.json` to ensure temporal traceability.
- **Objective:** To maintain an immutable history of external source responses, allowing for reprocessing if business rules or NLP requirements change.

5.2 Silver Layer

In this stage, the cleaning, normalization, and structuring of the data collected in the Bronze layer take place.

- **Processing:** Irrelevant fields are filtered, null values are handled, and initial NLP techniques (such as tokenization and stopword removal) are applied to reviews and comments.
- **Format:** Processed data is stored in parquet format, optimizing storage efficiency and query speed.
- **Tools:** Libraries such as Pandas or Polars will be utilized for efficient data structure manipulation.

5.3 Gold Layer

The final layer contains highly aggregated and enriched data, ready for consumption by the functional user (music analyst).

- **Analysis:** Final sentiment metrics (positive, negative, neutral) are calculated alongside statistical trend summaries.
- **Structure:** Tables are optimized for the interactive dashboard, allowing for the cross-referencing of artist popularity (API) with public perception (Scraping).

5.4 Orchestration

The management of the entire data flow will be centralized using Apache Airflow, ensuring the pipeline is reliable and scalable.

- **Ingestion (Bronze):** An Airflow DAG scheduled according to the defined periodicity will execute the extraction scripts.
- **Processing (Silver/Gold):** FileSensors will be implemented to detect the arrival of new files in the Bronze layer, automatically triggering downstream transformations.
- **Deployment:** The entire orchestration infrastructure and the Plotly Dash application will be managed via Docker containers to guarantee project portability.

6 Functional User

- The primary functional user of this dashboard is a music analyst or community manager. This user is interested in understanding public opinion about artists, albums, and songs.
- The user has an intermediate technical level. They do not need to know programming, but they can understand dashboards, charts, and basic data concepts.
- Their main need is to analyze trends, identify popular artists, and understand audience preferences. They use this information to support decisions such as marketing strategies, content creation, or music promotion.
- The decision-making context is based on real user behavior and opinions. For example, they may want to know which artist is becoming popular or how people react to a new album.
- This user benefits from the sentiment analysis pipeline because it transforms large amounts of unstructured data into clear insights.Reddit provides natural discussions and opinions, while Last.fm provides listening behavior data, such as most played artists and songs. [1]
- By combining both sources, the dashboard helps the user understand both what people say and what they listen to, improving decision-making.

7 User Stories

- As an Artist Manager, I want to visualize the average sentiment score of album reviews from Pitchfork and Reddit so that I can evaluate the initial public reaction to a new release.Market Trend Comparison (Storytelling):
- As a Music Label Analyst, I want to compare the listener statistics from Last.fm with the sentiment volume on social platforms so that I can identify which trending artists have the most loyal and positive fanbases.Data Quality Monitoring (Governance):
- As a Data Engineer, I want to monitor the status of the Bronze ingestion layer (Last.fm API) so that I can ensure the pipeline is consistently collecting raw JSON data without loss or corruption.Negative Feedback Identification (Storytelling):

- As a Digital Strategist, I want to extract and analyze recurring negative keywords in community discussions so that I can understand specific points of friction or dissatisfaction regarding a musical tour or event.Ingestion Reliability (Governance):
- As a Project Lead, I want to verify that the web scraping process respects the periodicity and structure of the target site so that the Silver layer receives high-quality, normalized text for NLP processing.

8 Conclusions

- In this project, we explored how to analyze public perception of music using data from Reddit and Last.fm.
- We found that Reddit is a valuable source of qualitative data, where users share opinions, recommendations, and discussions about music. This type of data is useful for sentiment analysis and understanding trends in a natural context.
- On the other hand, Last.fm provides structured data about listening behavior, such as artists, songs, and play counts. [1] This allows us to complement opinions with real user actions.
- By combining both sources, we can create a more complete view of music trends. This approach helps identify patterns between what people say and what they actually listen to.
- However, there are limitations, such as the size of the dataset and the fact that data comes from specific platforms. Despite this, the project demonstrates that social media and APIs are powerful tools for data analysis.
- Finally, this project shows how data pipelines and dashboards can support better decision-making in the music industry.

9 References

References

[1] Dataquest. (2024). Getting Music Data with the Last.fm API using Python. Available at: https://www.dataquest.io/blog/last-fm-api-python/

[2] Reddit. (n.d.). r/musicsuggestions subreddit. Available at: https://www.reddit.com/r/musicsuggestions/